

Requirement Analysis Phase-II

Technology Stack (Architecture & Stack)

Date	28-09-2026
Team ID	SWTID-2026-8620
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technology Architecture:

The architecture of our system follows a 3-tier ServiceNow architecture. The user uploads the spreadsheet via the User Interface layer. The Application Logic layer handles file validation, loading into Import Set Table (Staging), Transform Map execution, and coalesce logic. The Database layer stores data in Import Set Table and finally in target tables like CMDB, Incident, User. For error handling, failed records are moved to Error Log table with status.

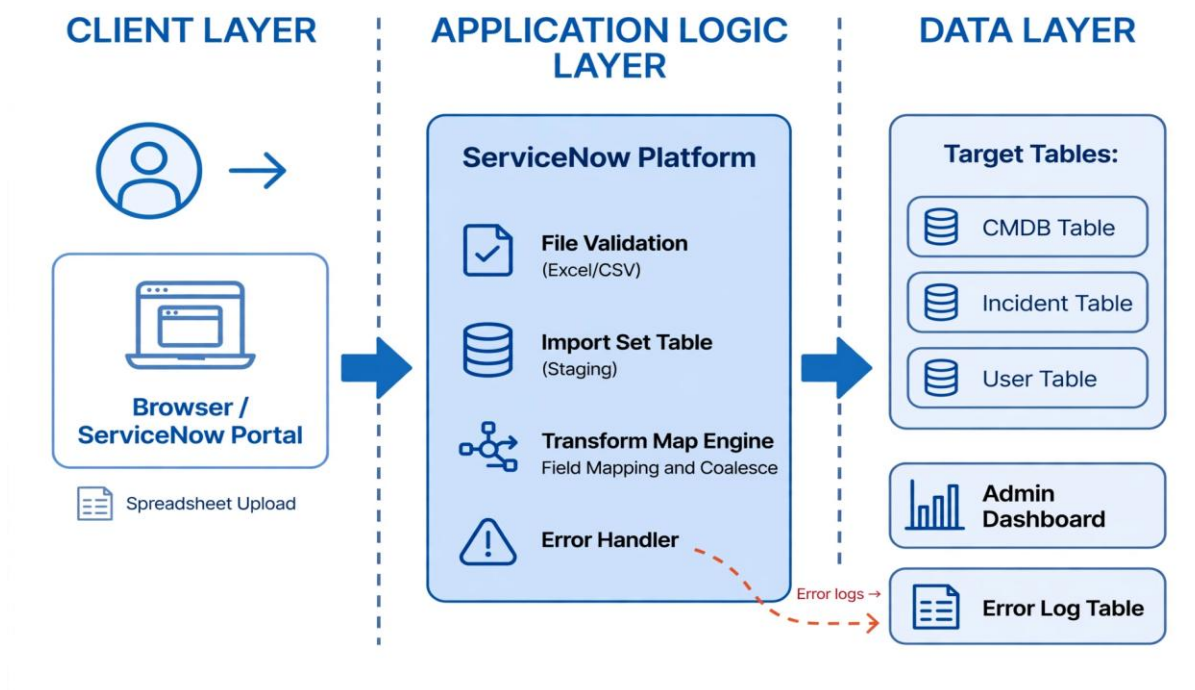


Table 1: Components & Technologies

S.No	Component	Technology / Tool	Description
1	User Interface	ServiceNow Service Portal, HTML, CSS, JavaScript	For uploading spreadsheet file and showing import status
2	Application Logic	ServiceNow Transform Maps, Business Rules, GlideScript	For validation, mapping, transformation logic
3	Database Storage	ServiceNow Import Set Table, CMDB / Incident Table	Staging table and final target tables
4	File Processing	Apache POI, ServiceNow Data Import API	For reading Excel/CSV and parsing data
5	Authentication	ServiceNow Role-based Access (Admin Role)	Only admin can import
6	Reporting	ServiceNow Dashboard & Reports	To show success/failure count and history
7	Deployment	ServiceNow Instance (Cloud)	Deployed on ServiceNow cloud

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	ServiceNow platform components	ServiceNow JavaScript
2	Security Implementations	Role-based access, Data validation, Encryption	ACLs • Business Rules • SSL
3	Scalable Architecture	Supports multiple target tables & large files	3-Tier Architecture
4	Availability	System available 99.9%	ServiceNow Cloud Infrastructure
5	Performance	Fast import of 10k+ rows	Glide Engine • Import Sets
6	Data Retention	Logs stored for 30 days for auditing	Table Archiving